Terzaghi consolidation: equations for the poroelastic check

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Problem and assumptions

A saturated column of height H, loaded at once by a constant compressive stress p_0 on its top face, consolidates as fluid leaves through the top only. The axisymmetric cylinder ($r_i = 0$, height H) is laterally confined, so the deformation is uniaxial along z, with $z = 0$ at the bottom and $z = H$ at the top.

- The constituents are incompressible: Biot coefficient equal to 1, no storage term.
- The permeability k is constant (Darcy law, $w = -k\,\partial p/\partial z$).
- The strain is small: the load is small compared with the constrained modulus M, so the response is linear.
- Bottom: $u_z = 0$ and no flow. Lateral surface: $u_r = 0$ and no flow. Top: drained ($p = 0$) and loaded by p_0.

Governing equations

The pore pressure obeys a diffusion equation with coefficient $c_v = kM$. It follows from four relations, with the effective stress σ' taken positive in tension.

Momentum balance (no body force) in one dimension. The total stress is uniform and equal to the applied one:

$$\frac{\partial}{\partial z}\left(\sigma'_{zz} - p\right) = 0 \quad \Rightarrow \quad \sigma'_{zz} - p = -p_0.$$

Skeleton law, with the constrained (oedometric) modulus M:

$$\sigma'_{zz} = M\,\varepsilon_{zz}, \qquad \varepsilon_{zz} = \frac{\partial u_z}{\partial z}.$$

Darcy law and mass balance for incompressible constituents (the volume change equals minus the divergence of the flux):

$$w = -k\,\frac{\partial p}{\partial z}, \qquad \frac{\partial \varepsilon_{zz}}{\partial t} = -\frac{\partial w}{\partial z} = k\,\frac{\partial^2 p}{\partial z^2}.$$

Since $M\,\varepsilon_{zz} = p - p_0$, one has $\partial \varepsilon_{zz}/\partial t = (1/M)\,\partial p/\partial t$, and the consolidation equation is

$$\frac{\partial p}{\partial t} = c_v\,\frac{\partial^2 p}{\partial z^2}, \qquad c_v = k\,M.$$

Conditions and solution

The load is carried entirely by the fluid at $t = 0^+$, then transferred to the skeleton as the pressure dissipates through the drained top:

$$p(z, 0^+) = p_0, \qquad p(H, t) = 0, \qquad \frac{\partial p}{\partial z}(0, t) = 0.$$

With the dimensionless time $T = c_v t/H^2$ and $M_m = (2m+1)\pi/2$, the series solution for the pressure is

$$\frac{p(z,t)}{p_0} = \sum_{m=0}^{\infty} \frac{2(-1)^m}{M_m}\,\cos\!\left(\frac{M_m z}{H}\right) e^{-M_m^2 T}.$$

The bottom is fixed, so the top displacement is the integral of $\varepsilon_{zz} = (p - p_0)/M$ over the height. It gives the degree of consolidation $U(T)$, which goes from 0 at $T = 0$ to 1 for $T \to \infty$:

$$u_z(H,t) = -\frac{p_0 H}{M}\,U(T), \qquad U(T) = 1 - \sum_{m=0}^{\infty} \frac{2}{M_m^2}\,e^{-M_m^2 T}.$$

The final settlement is $-p_0 H/M$. The series converge fast for T above about 0.01 and slowly at very small T, where several hundred terms are needed (500 are used in the check).

Parameters and compared quantities

The constrained modulus comes from the linearisation of the Mooney-Rivlin law about the reference state,

$$W = C_{10}\,(\bar{I}_1 - 3) + C_{01}\,(\bar{I}_2 - 3) + \frac{(J-1)^2}{D}$$

(convention to be confirmed against the framework), which gives

$$\mu = 2\,(C_{10} + C_{01}), \qquad K = \frac{2}{D}, \qquad M = K + \frac{4}{3}\,\mu.$$

The load must keep the strain p_0/M small (below about 1 percent), so that the nonlinear solution stays close to the linear one. The time scale is $t_c = H^2/c_v$, with k in the units of the Darcy law of the framework (length to the fourth power per force and time).

Quantity	Simulation	Analytical
$p(0,t)/p_0$	fluid pressure averaged over the bottom face	series for $z = 0$
$U(T)$	$-u_z(H,t)\,M/(p_0 H)$, with u_z averaged over the top face	$1 - \sum_m \frac{2}{M_m^2} e^{-M_m^2 T}$

Both quantities are compared as functions of T, so the curves do not depend on k, M or H.